

Preserving engineering assessments across fresh language-model sessions: a paired EAL/2 workflow pilot

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Abstract

Engineering decisions often depend on dated evidence whose applicability changes between work sessions. A language model inheriting a previous answer must distinguish a reusable observation from a conclusion that has become stale. We evaluate a workflow built around EAL/2, an executable engineering argument language, against an ordinary specification-and-notes workflow. A retained pilot comprises six synthetic tasks, two donor models, two recipient models, two recipient tool settings and four repetitions: 192 paired blocks, 384 sequences and 4,224 fresh sessions. Each sequence contains one donor and ten recipients. The EAL/2 host supplies a checked assessment and manages observation reuse; the ordinary workflow receives the latest answer and can request evidence when native tools are enabled. The comparison therefore concerns these complete workflows. Post-collection AI-assisted coding leaves 31 answers ambiguous. Among 1,920 recipient answers per arm, correctness relative to the task reference is bounded at 97.71–99.11% for EAL/2 and 38.85–38.91% for ordinary notes. Cumulative input plus output tokens, including donors, are 1,943,638 and 2,992,739 respectively, a 35.05% reduction. These descriptive results are conditional on the selected tasks and unvalidated AI coding. They neither isolate a language-notation effect nor establish performance on real engineering projects. The replication package preserves prompts, answers, evidence events, accounting and uncertainty, and identifies the annotation and design work required before a confirmatory evaluation.

Keywords: Executable arguments, engineering evidence, large language models, context transfer, empirical software engineering, reproducibility

1. Introduction

An engineering assessment can be correct when written and misleading when reused. A test result may concern an earlier release; a calibration may cease to apply after a configuration change; an observation may exceed its permitted age. A successor working session needs both the decision rule and evidence that is eligible for its current target. Retaining a persuasive explanation does not by itself preserve those conditions.

Language-model workflows make this problem concrete. A fresh session can inherit a specification and a predecessor’s notes without inheriting the provider’s conversation state. The notes may communicate a useful conclusion, but they also create a risk of repeating that conclusion after its evidential basis changes. A different design retains structured claims, methods, evidence requirements and observations in the host. The host can check applicability and compute a task result before asking the model to explain it. This design spends engineering effort on explicit rules and trusted integrations, and its benefit must be measured against that operational cost.

This paper reports a completed pilot of the latter design using EAL/2. The experiment tests a bounded practical question: on six authored engineering tasks, how do recipient decision correctness and cumulative model-token use differ between a host-assessed EAL/2 workflow and a specification-and-notes workflow across ten fresh recipient sessions? It also asks how those outcomes vary with task, session position, recipient model and native tool availability. The result is a comparison of complete workflow assignments. The EAL/2 arm receives an

automatically prepared current assessment that the ordinary arm does not receive automatically. Equal information delivery is therefore not part of the treatment contrast.

The contributions are a precisely specified repeated-session comparison, a retained record connecting answer measurements to acquisition and API accounting, and an analysis that keeps ambiguous answers in the denominator. The paper distinguishes three forms of incompleteness: the paid collection is complete; decision coding is partly unresolved and lacks independent human validation; the planned prospective allocation remains blocked. Treating these as separate statuses prevents a successful workflow run from being mistaken for a completed scientific evaluation.

2. Related work and the scope of the comparison

External execution is an established way to complement language generation. Program-aided language models generate programs and delegate computation to an interpreter [1]; Logic-LM combines language models with symbolic solvers [2]. ReAct interleaves reasoning and actions to obtain information and act on an environment [3]. These approaches motivate separating language production from executable operations. The present intervention instead supplies a pre-authored, source-bound engineering rule and persistent evidence management. It does not evaluate model-generated programs, solver selection or general planning ability.

Structured assurance representations distinguish claims, evidence and supporting argument structure; SACM provides a metamodel for such representations [4]. EAL/2 is a particular executable design whose host binds acquisition and reasoning methods. This study makes no SACM-conformance claim and does not measure the quality of a complete safety or assurance case. Its outcome is a small, explicitly defined readiness judgement under synthetic evidence changes.

Long-context studies show that the presence of information in a prompt does not ensure that a model will use it effectively [5]. Here, however, every recipient starts with a fresh provider session. The study concerns what a host prepares and retains between sessions, rather than the position of evidence in one long conversation. Likewise, research on LLM-based judges documents useful agreement and systematic biases under particular benchmarks [6]. It supplies a reason to examine assessor error; it does not validate the AI-assisted coding used here. Our measurement procedure, unresolved cases and absence of independent assessor validation are reported explicitly.

3. Workflow and task semantics

3.1. What persists between sessions

Both arms receive the same task specification and may retain project notes. The runner stores the latest answer verbatim as a handoff note, subject to an 8,192-byte limit. The next session is a new model invocation without provider conversation history or a previous-response link. Additional calls within a session can complete tool use, but they do not connect that session to its predecessor’s provider state.

In the ordinary arm, context preparation contributes no automatic evidence assessment. The recipient receives the specification, retained material and current task time. When native tools are enabled it can request a dated snapshot through the `probe` tool. Tool availability is an opportunity to acquire evidence, not a guarantee that acquisition occurs. When tools are disabled, the model relies on the supplied and retained text.

In the EAL/2 arm, a trusted host assesses the retained EAL source before model invocation. It checks observation eligibility, collects evidence when necessary, reuses eligible observations and supplies the resulting task context. The model is not given the raw EAL source as its principal task input. Consequently, the experiment evaluates the behaviour of a host that interprets EAL/2 and prepares model context. It is not a test of whether either model understands EAL/2

syntax. Native model tools, when enabled, remain available in addition to this automatic host activity.

The intervention combines explicit source, deterministic task evaluation, persistent observations, evidence acquisition policy and prepared model context. A difference in outcomes may arise from any combination of these components. In particular, automatic delivery of a current result can account for an advantage even if notation has no independent effect. The disabled-native-tools condition isolates neither the host nor external computation, because the EAL host still acquires and assesses evidence there.

3.2. Eligibility, three-valued readiness and source binding

Each task specifies requirements over dated facts. A fact is eligible when it is active, matches the requirement’s scope keys and has an age between zero and the requirement’s maximum, inclusively. The uniquely newest eligible fact is selected. An absent or ineligible fact gives an unknown requirement value. The selected corpus contains no unresolved ties between simultaneously newest facts.

For conjunctive tasks, any false requirement makes the conjunction false; otherwise an unknown requirement makes it unknown. Alternative routes are evaluated as conjunctions: one true route suffices, all refuted routes give false, and the remaining cases are unknown. The task adapter maps true, false and unknown to **ready**, **not_ready** and **undetermined**. Scope may bind different requirements to different version keys, so a firmware change need not invalidate an unrelated calibration unless that requirement’s declared scope requires it.

The snapshot contract permits reuse for 60 seconds inclusively. Sessions advance the synthetic task clock by one minute. Evidence and scope changes occur at even-numbered positions; the following odd position retains the same snapshot under this policy. These are authored validity assumptions. They do not establish that a real engineering observation remains valid for one minute, nor that absence of a new observation means the physical system is unchanged.

The retained source binds evidence to a task identifier, a rule digest and a target-identity digest. Its reasoning selector is `method "experiment/task-rules/1"`. The example in Listing 1 is an excerpt from the actual cutover source, formatted for publication. The complete source, including the evidence digests and claim statement, is supplied in the replication package and checked by the repository parser.

```

1 reasoning task_rules {
2   method "experiment/task-rules/1";
3   rationale "Evaluate eligible dated facts using the source-bound three-valued task rule; negative
      and unknown outcomes are completed calculations.";
4   require "snapshot_minute" >= 0;
5 }
6
7 argument result {
8   conclusion criterion_evaluated;
9   reasoning task_rules;
10  evidence report;
11 }

```

Listing 1: Reasoning and argument declarations from the retained cutover source. The complete, parser-validated source is included with the article.

The claim identifier `criterion_evaluated` refers to successful completion of the declared calculation. A supported calculation can yield negative or unknown readiness. The host’s assessment status must therefore not be substituted for the task’s readiness output. This distinction is material to the experiment: successful execution of a rule does not establish that the engineering action should proceed.

3.3. The reference computation

The task reference is implemented separately from the EAL interpreter and its installed method. It derives readiness from the recorded rule, scope and timeline, and checks the authored

Table 1: The six selected cases. Every case has eleven task-time positions and ten recipient assessments per sequence.

Case	Rule family	Engineering distinction exercised
Release	Conjunction	Tests, signature and rollback obligations; refutation and expiry.
Cutover	Conjunction	Schema compatibility, restored backup and bounded replica lag.
Read routes	Alternatives	Alternative database read routes, including unknown and refuted routes.
Dispatch	Alternatives	Direct or tunnel route; a valid alternative can preserve readiness.
Firmware	Applicability	Selective binding to firmware, calibration and policy versions.
Version bindings	Applicability	Multiple version bindings in a migration assessment.

expected decisions. This provides an executable reference for this corpus. It is not externally validated engineering ground truth: the task authors supplied the rules and synthetic facts shared by both implementations. The article’s reproduction script independently recomputes all reference decisions from those retained facts and reconciles them with every saved score. Agreement checks catch divergent implementations, but cannot establish that the common task specification represents an actual engineering decision adequately.

4. Experimental method

4.1. Questions, units and selected cases

The primary endpoints are recipient decision correctness and cumulative input plus output tokens through recipient horizon ten. The questions are whether the EAL/2 workflow preserves a correct current decision, whether its cumulative model-token use is lower after including the donor, and what qualification the unresolved measurement and allocation status place on those findings. Task and configuration profiles are descriptive diagnostics rather than separate confirmatory hypotheses.

The selected pool contains six separately AI-authored synthetic cases in three families (Table 1). Eight threshold-calibration cases are excluded from the principal estimand. The selected tasks were not randomly sampled from projects, organisations or engineering domains. Each trajectory exercises changes in support, refutation, expiry or applicability. An authored sequence of expected decisions is a corpus property; repeated model sessions do not create new independent tasks.

The design crosses six cases, two donor models, two recipient models, two recipient native-tool settings and four repetitions. Each resulting block contains both workflows, giving 192 paired blocks and 384 sequences. Every sequence comprises one donor followed by ten recipients, for 4,224 sessions. The 48 case–donor–recipient–tool strata each contain four paired repetitions. All donor sessions have native tools enabled, including those in blocks whose recipients have tools disabled.

The paired block is the unit for comparisons across repeated stochastic executions. A sequence is the unit for within-arm recipient averages; its ten recipient outcomes share inherited material and are dependent. The analysis does not treat 3,840 recipient answers as independent Bernoulli trials. The fixed pool and balanced allocation give equal weight to the six tasks and the model/tool configurations. That weighting defines this experiment’s estimand, not a population distribution of engineering work.

4.2. Models, execution and the frozen contract

The model labelled **plain** is **gpt-4.1-nano-2025-04-14**. The model labelled **reasoning** is **gpt-5-nano-2025-08-07**, with reasoning effort set to **low**. The former has no explicit reasoning-effort setting. These labels distinguish two model configurations; changing between

Table 2: Execution and processing identities retained by the replication package.

Item	Retained identity or status
Collection	GitHub Actions run 36477862213; revision d52e2bd0898ed519253aedbbc8e002e493dd64d2
Frozen study	eal-model-transfer-pilot-6; pilot role; protocol 6.0.0
Source language	EAL/2; distinct from package and protocol versions
Annotation export	Corrected export run 36487342999; artefact 10998924773
Offline finishing	Run 36492423435; revision 31a3cba769948e123a59ca35c54c5135ad005f54
Finished artefact	11001997190; analysis available, prospective allocation blocked

them also changes the model family and snapshot. They cannot identify a pure causal effect of reasoning effort.

The frozen plan uses seed 20260928. The runner constructs the crossed blocks, shuffles their order and randomises arm order within each block. Five workers process independent blocks concurrently; the two arms within a block run sequentially in their assigned order. Sessions within an arm retain their task-time order. Random ordering reduces a systematic association between workflow and execution order, but a shared provider batch can still introduce dependencies and service variation.

Responses are ordinary prose. Limits are three API calls per session, 4,096 output tokens per call, 32,768 request bytes and a 90-second request timeout. The retained plan has a USD 2 budget, a 21,600-second total runtime limit and five workers. The collection job’s segment allowance is 6,000 seconds. The actual retained plan and execution contract govern this collection; an older 7,200-second value in the descriptive protocol is not used as its realised runtime limit. This discrepancy is preserved in the audit record rather than silently normalised.

Collection ran on 28 September 2026 from 20:20:10 to 21:08:04 UTC, with a recorded segment duration of 2,874.54 seconds. The frozen run identity is 984bd05d-6366-41c3-b9ae-ee8766835b08. The successful offline finishing workflow imported AI labels and produced analysis and planning reports without new collection calls. Table 2 distinguishes collection from processing. A processing revision does not retroactively change the implementation that produced the observations.

4.3. Post-collection decision coding

Prose answers require measurement before comparison with the task reference. The retained answers were exported with workflow metadata masked. Following collection, AI-assisted coding assigned a stated-decision category using explicit decision phrases and semantic adjudication. The recorded procedure covers all 4,224 answers through AI-authored rules and 222 adjudication entries. It is not 4,224 independent model judgements. The coding record identifies OpenAI ChatGPT/Codex; the exact serving model identifier was not exposed.

Coding asks what decision the answer states, not whether the underlying decision is correct. The latter comparison is made against the reference computation. A plausible explanation does not repair an explicitly wrong final category, and a matching category does not demonstrate that the explanation is grounded or internally consistent. This paper therefore reports *decision correctness relative to the task reference*; it does not report validated explanation quality.

The exported text can reveal the workflow through its wording, and the AI assessor had prior task context. Metadata masking is consequently weaker than assessor independence or effective blinding. There is no independent human validation, inter-rater reliability estimate or measured assessor error rate. The AI coding is a disclosed post-collection measurement amendment, not a preregistered independently validated assay. Frozen collection settings provide reproducibility, but do not establish an externally registered protocol.

The final categories comprise 2,345 `ready`, 850 `not_ready`, 998 `undetermined` and 31 `ambiguous` answers. Ambiguous cases retain unknown correctness. Of these, 28 are recipient

answers and three are donors. No ambiguous answer is discarded or assigned a convenient reference-matching label. The package includes answer text, selected quotations, adjudication notes, assessor provenance and checksums sufficient to locate the measurements for independent review.

4.4. Estimands and uncertainty

Let Y_{baj} indicate that the coded decision matches the reference in block b , arm a and recipient position j . For $B = 192$, $H = 10$ and $a \in \{E, O\}$, define

$$\hat{Q}_a = \frac{1}{B} \sum_{b=1}^B \frac{1}{H} \sum_{j=1}^H Y_{baj}, \quad \hat{\Delta} = \hat{Q}_E - \hat{Q}_O. \quad (1)$$

The target $Q_a = \mathbb{E}[\hat{Q}_a]$ averages repeated executions of this fixed task and configuration allocation, and $\Delta = Q_E - Q_O$. Each ten-recipient sequence contributes one block mean. Write q_a^{obs} and $\delta^{\text{obs}} = q_E^{\text{obs}} - q_O^{\text{obs}}$ for the fully resolved realised values of the estimators. For unknown correctness, set Y to zero for the lower endpoint and one for the upper endpoint. With C_a correct answers, U_a unknown answers and $N_a = BH$, the descriptive envelopes are

$$q_a^{\text{obs}} \in \left[\frac{C_a}{N_a}, \frac{C_a + U_a}{N_a} \right], \quad \delta^{\text{obs}} \in \left[\frac{C_E - C_O - U_O}{BH}, \frac{C_E + U_E - C_O}{BH} \right]. \quad (2)$$

These bounds cover the unresolved categories conditional on the other codes being correct. They are not confidence intervals and do not cover assessor errors among apparently resolved answers.

Let T_{baj} be recorded input plus output tokens, with $j = 0$ denoting the donor. The resource endpoint is the ratio of aggregate totals:

$$\hat{R}_T = 1 - \frac{\sum_b \sum_{j=0}^H T_{bEj}}{\sum_b \sum_{j=0}^H T_{bOj}}. \quad (3)$$

The corresponding target R_T is one minus the ratio of expected aggregate token totals under repeated executions of this allocation. Equation 3 weights per-pair reductions by ordinary-arm token use. Cached input is already included in input tokens; reasoning tokens are already included in output tokens. Neither is added again. API attempts, cost estimates, native probe calls, host acquisitions, observation reuses and elapsed times are separate quantities. A token saving does not imply fewer evidence acquisitions or lower total engineering cost.

The frozen engineering criteria require $\Delta > -0.05$, $Q_E > 0.90$ and $R_T > 0.20$ at $H = 10$. The margins are prospective design choices, not validated developer-utility thresholds. The saved analysis also supplies conditional sampling intervals. For correctness it uses an empirical Bernstein bound on whole paired trajectories, allowing independent non-identically distributed bounded observations [7]. Unknown answers enter through a conservative variance envelope. For resources it uses an approximate stratified paired delta-method interval with a t correction [8]. Appendix A records the implementation and assumptions.

The displayed two-sided intervals allocate a total nominal error probability of 0.05 across three endpoints. The joint engineering decision instead uses an intersection-union of three one-sided tests, each at 0.05. Their three lower bounds must not be presented as jointly 95% simultaneous lower bounds. Both calculations depend on the measurement and execution assumptions; neither turns a pilot into a confirmatory evaluation.

Figure 1 illustrates these operations with the retained pilot values. In panel (a), $\bar{l}$ and $\bar{u}$ are the means of the lower and upper paired-block differences, and the Bernstein radius r expands their observed coding envelope to a sampling interval for Δ . Panel (b) divides both token totals by the ordinary total; the remaining span to one gives $\hat{R}_T$.

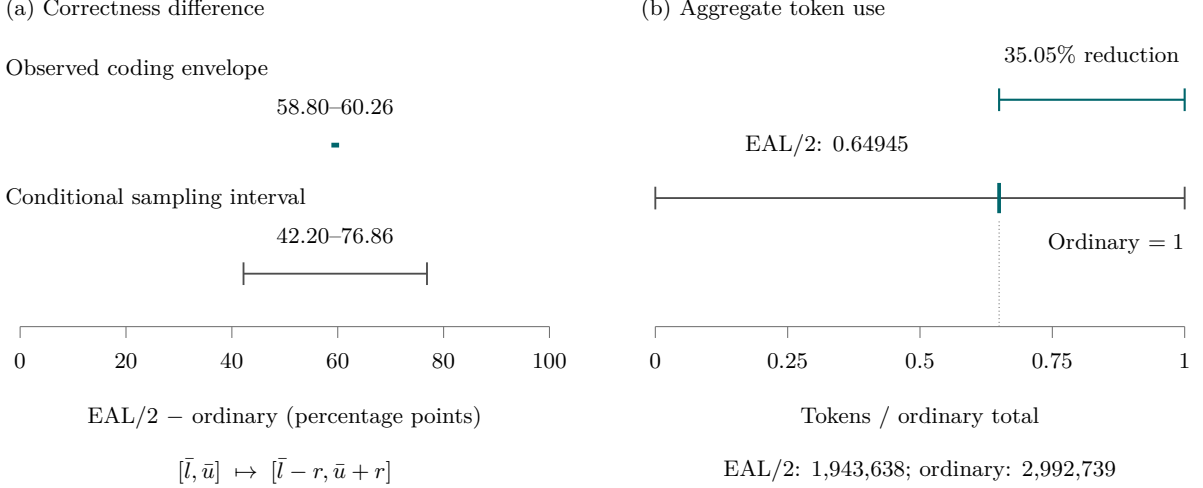


Figure 1: Observed uncertainty and aggregate token use at $H = 10$. (a) The thick segment covers unresolved coding; the thin capped segment adds the two-sided Bernstein allowance for expected correctness difference. It uses $\alpha_t = 0.05/6$ and assumes independent blocks and correct resolved codes. (b) Ordinary token use defines one; the span above the EAL/2 ratio is the token reduction. Both totals include donors and ten recipients.

5. Results

5.1. Collection completeness and decision correctness

All 384 planned sequences and 4,224 sessions are recorded. The article reconstruction reconciles every session with its API attempt identifiers, recomputes the task reference from the retained corpus and checks all saved correctness scores. All 5,453 API attempts have recorded usage and cost estimates. Execution completeness therefore does not depend on imputing missing calls or treating absent usage as zero.

For the 1,920 recipients in the EAL/2 arm, 1,876 coded decisions match the reference, 17 do not and 27 remain unknown. The corresponding ordinary-arm counts are 746, 1,173 and one. Equation 2 gives correctness envelopes of 97.71–99.11% and 38.85–38.91%, respectively, and a difference of 58.80–60.26 percentage points. These are large descriptive differences for the selected workflow assignments, conditional on the resolved AI codes.

The direction is consistent across all six tasks (Table 3 and Figure 2). The lowest EAL/2 lower endpoint is 95.6% for Firmware; ordinary-arm endpoints range from 33.1% for Release to 48.4% for Read routes. The shape of the task-time profiles is more informative than an overall average alone (Figure 3). Recipients encounter authored changes as well as intervals of permitted reuse. A correct predecessor answer can continue to match the reference for one position and then become stale. The plotted profiles describe this sequence dependence; they do not estimate an independent time effect.

5.2. Recipient models and native tool availability

Table 4 disaggregates recipients by model and native tool setting, pooling donors and tasks. Every cell contains 480 recipient answers. The highest observed EAL/2 correctness occurs for GPT-4.1 nano with native tools disabled: all 480 decisions match the reference. In this condition the EAL host still supplies its checked assessment. This observation therefore cannot support a claim that external evidence or computation is unnecessary.

Ordinary-workflow correctness is similar across these configurations, despite native tool availability in half of them. An available probe need not be invoked at every session, and an invocation need not produce a correctly stated final decision. The aggregate record contains 488 ordinary recipient native probe calls across all 1,920 recipients, including the disabled-tool cells, versus 470 for EAL/2. The data support inspection of acquisition behaviour, but do not identify

Table 3: Recipient outcomes by case; each arm has 320 answers per case. C/I/U means correct/incorrect/unknown. Ranges assign unknown correctness adversely and favourably; they are not confidence intervals.

Case	EAL/2		Ordinary notes	
	C/I/U	Correct (%)	C/I/U	Correct (%)
Release	309/6/5	96.6–98.1	106/214/0	33.1
Cutover	316/2/2	98.8–99.4	109/211/0	34.1
Read routes	318/1/1	99.4–99.7	155/165/0	48.4
Dispatch	310/4/6	96.9–98.8	138/181/1	43.1–43.4
Firmware	306/4/10	95.6–98.8	129/191/0	40.3
Version bindings	317/0/3	99.1–100.0	109/211/0	34.1

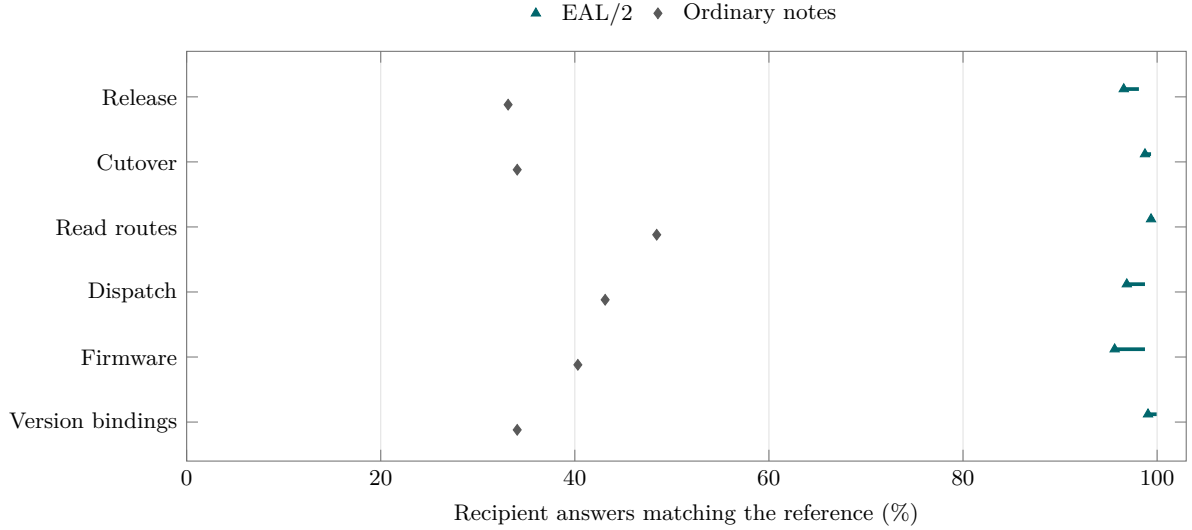


Figure 2: Decision-correctness envelopes for the six selected cases, pooling donor and recipient configurations. Markers show the lower endpoint; short horizontal segments extend to the upper endpoint when answers are ambiguous. Each marker represents 320 recipient answers nested within 32 sequences. The ranges express unresolved coding, not sampling precision.

a causal effect of actually using the probe: tool use is chosen during the assigned workflow and is a post-assignment variable. A comparison restricted to callers would select on model behaviour.

5.3. Cumulative model resources and host work

The EAL/2 workflow uses 1,943,638 input plus output tokens; ordinary notes use 2,992,739. The resulting aggregate reduction is 35.05%. Table 5 includes the donor and separates model usage from collector activity. EAL/2 has lower aggregate input and output counts, although the reductions differ. Figure 5 shows how those differences accumulate through the recipient horizon.

EAL/2 uses fewer cumulative tokens in 191 of the 192 paired blocks. Figure 4 retains the distribution rather than replacing it with the aggregate ratio. It also makes clear that a ratio of totals weights blocks by ordinary-arm token use, whereas an unweighted mean of pair ratios answers a different question. No pair is excluded for having an unfavourable ratio.

Lower model-token use accompanies additional host evidence work. EAL/2 records 1,152 automatic host acquisitions, 960 host reuses and 562 native probe calls, giving 1,714 collector calls in total. Ordinary notes record 667 native probe calls and no automatic host acquisitions. A host reuse is not a new collector call. This is a model-token saving under a workflow that performs more recorded acquisitions, not an across-the-board reduction in computational activity.

The retained API cost estimates are USD 0.29840445 for EAL/2 and USD 0.39637595 for ordinary notes, totalling USD 0.69478040 within the frozen USD 2 ceiling. Pricing, caching and

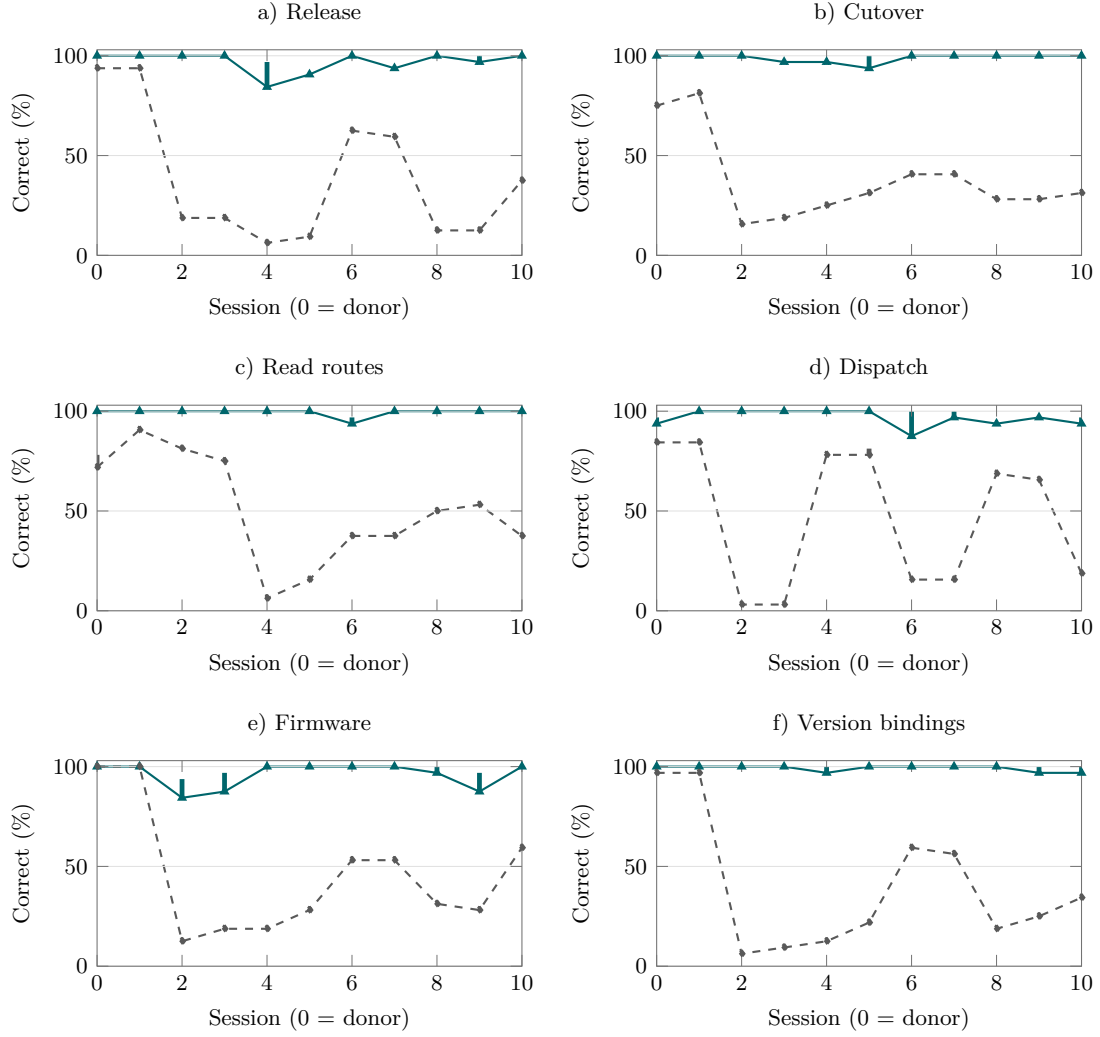


Figure 3: Correctness by task and session position. Solid teal triangles denote EAL/2 and dashed grey diamonds ordinary notes; vertical segments cover ambiguous coding. Each position contains 32 answers per arm. Position zero is the donor and is excluded from recipient correctness; positions one to ten are fresh recipients. The task fixtures change at even positions, with the next odd position retaining an eligible snapshot. The horizontal connections show ordered profiles, not independent observations or interpolation of task state.

Table 4: Recipient configuration profiles. Native tools refer to the recipient’s optional probe, not the EAL host’s automatic evidence activity. All donors have native tools enabled.

Tools	Recipient	Workflow	C	I	U	Correct (%)
Off	4.1 nano	EAL/2	480	0	0	100.0
Off	4.1 nano	Ordinary	189	291	0	39.4
Off	5 nano (low)	EAL/2	475	0	5	99.0–100.0
Off	5 nano (low)	Ordinary	181	299	0	37.7
On	4.1 nano	EAL/2	447	16	17	93.1–96.7
On	4.1 nano	Ordinary	187	292	1	39.0–39.2
On	5 nano (low)	EAL/2	474	1	5	98.8–99.8
On	5 nano (low)	Ordinary	189	291	0	39.4

Table 5: Recorded resources. Initial denotes 192 donors per arm; recipients denotes 1,920 subsequent sessions. Token totals include all linked API attempts. Cost is the retained plan’s API estimate, not an invoice or total project cost.

Measure	EAL/2			Ordinary notes		
	Initial	Recipients	Total	Initial	Recipients	Total
Input tokens	178,630	1,340,792	1,519,422	351,933	2,145,471	2,497,404
Output tokens	35,425	388,791	424,216	40,578	454,757	495,335
Input + output tokens	214,055	1,729,583	1,943,638	392,511	2,600,228	2,992,739
API attempts	284	2,390	2,674	371	2,408	2,779
Native probe calls	92	470	562	179	488	667
Host acquisitions	192	960	1,152	0	0	0
Host observation reuses	0	960	960	0	0	0
Estimated API cost (USD)	0.029616	0.268788	0.298404	0.042560	0.353816	0.396376

different model input rates mean that money and token counts are not interchangeable endpoints. The recorded collection segment lasts 47.91 minutes with five workers. Summed session elapsed times, about 5,290 and 5,406 seconds by arm, overlap in wall time and include more than model inference. They must not be added and interpreted as a serial workflow duration. Source authoring, method development, annotation, infrastructure and human review costs were not measured.

5.4. Conditional intervals and the blocked prospective allocation

The retained simultaneous-interval calculation yields 42.20–76.86 percentage points for the correctness difference, 89.63–100% for EAL/2 correctness, and 33.64–36.47% for cumulative token reduction. The first two are conservative bounded-outcome intervals; the third is an approximate resource interval. They condition on the fixed cases, independent paired execution assumptions and correct coding of resolved answers. They do not incorporate an estimated AI-assessor error distribution.

The separate one-sided decision calculation gives lower bounds of 47.16 percentage points, 92.07% and 34.10%, respectively. The saved report marks its joint engineering criteria as met while retaining `pilot_only` status. The fact that the two-sided absolute-correctness interval reaches below 90% is compatible with this result: the intervals and one-sided decision use different tail allocations. It should not be hidden by reporting only the more favourable calculation.

The prospective planner returns `no_supported_allocation`. It requires complete pilot recipient annotation and at least four complete paired repetitions in each stratum for nuisance estimation. Although four pairs were collected in every stratum, ambiguous recipient answers reduce the minimum complete-pair count to one. All 48 strata have resource variance information, but this does not supply missing correctness information. No confirmatory plan was emitted and no evaluation collection was run. A successful offline pipeline therefore means that processing and reporting completed, not that the experiment satisfied every scientific prerequisite.

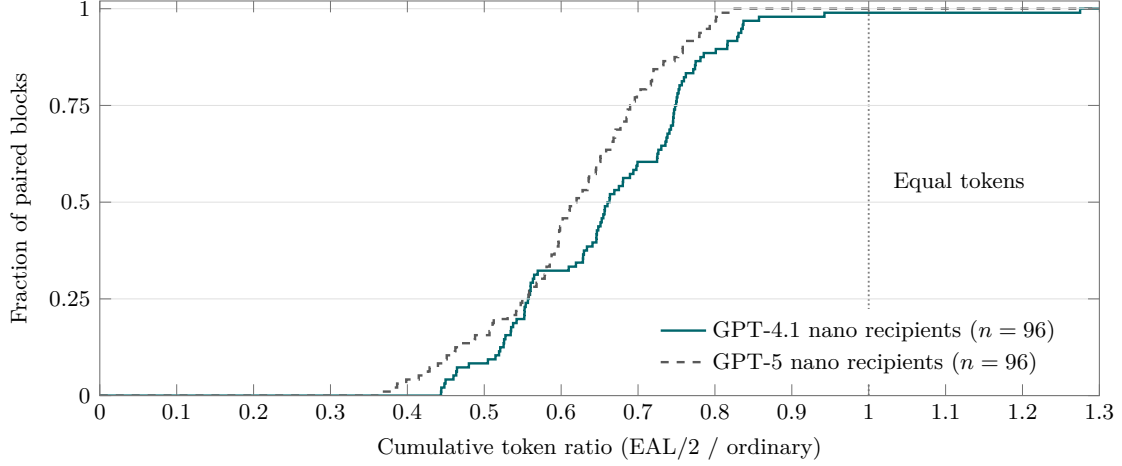


Figure 4: Empirical distributions of the paired cumulative token ratio, including donor and ten recipients. Each curve contains 96 paired blocks grouped by recipient model, pooling donor, task and tool settings. Ratios below one favour EAL/2. These are distributions of paired resource observations, not confidence distributions for the aggregate ratio.

5.5. Sensitivity to unresolved assessor validity

An unknown-answer envelope does not protect against misclassification of a resolved answer. To make this distinction explicit, Figure 6 presents an exploratory, post-collection sensitivity calculation. Let ϵ_E and ϵ_O denote adverse coding changes as fractions of all recipient answers in the respective arms. They are assumed error allowances, not measured error rates. A conservative descriptive lower gap is

$$L_{\Delta}(\epsilon_E, \epsilon_O) = 0.5880208 - \epsilon_E - \epsilon_O, \quad L_E(\epsilon_E) = 0.9770833 - \epsilon_E. \quad (4)$$

The positive-gap condition holds when the two allowances sum to less than 58.80 percentage points. The stronger absolute requirement $L_E > 0.90$ allows less than 7.71 percentage points of adverse EAL/2 coding change. These statements concern robustness of the observed envelopes only; sampling uncertainty and possible common specification error remain separate. The plot cannot substitute for independent annotation validation.

6. Discussion

6.1. What the pilot establishes

For these six synthetic tasks, the completed EAL/2 workflow produces many more reference-matching recipient decisions and uses fewer cumulative model tokens than the implemented ordinary-notes workflow. The result survives assigning all unresolved EAL/2 recipient answers incorrectly and the single unresolved ordinary recipient answer correctly. Retained acquisition events also show that the EAL host implements the intended alternation between collecting and reusing dated snapshots. These are measured properties of this particular execution, subject to the declared coding limitation.

The result is consistent with a practical design in which a host owns evidence eligibility and deterministic calculation, while a model communicates the result. It does not establish that the model has acquired a more reliable reasoning capability. A short checked context can reduce repeated prompt material, and automatic provision of the current answer can improve decision accuracy. Those are useful workflow properties if their upstream engineering and validation costs are acceptable. The experiment has not measured those costs or compared equally informative alternatives.

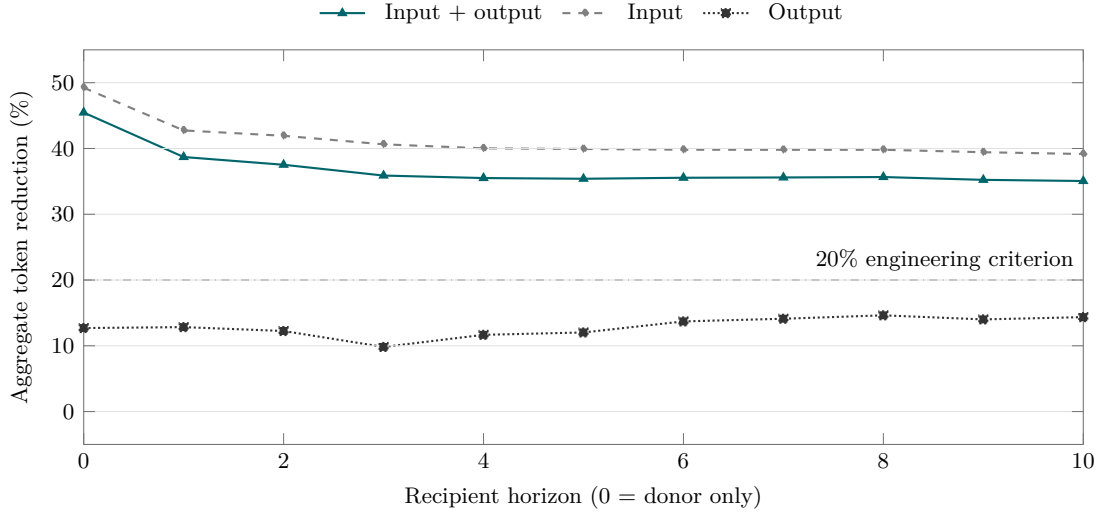


Figure 5: Observed aggregate token reductions as recipients accumulate, always including donors. Zero denotes donors alone. Input and output curves are components; the total curve uses their summed counts. The horizontal 20% reference is the frozen engineering criterion for the total at horizon ten. It is not a fitted effect, a cost threshold or a confidence limit.

6.2. Alternative explanations and a discriminating follow-up

The largest identification gap is the bundled treatment. Ordinary recipients sometimes have only the previous answer, whereas EAL recipients automatically receive a current checked assessment. A direct follow-up should independently vary delivery of the same current facts, provision of a computed decision, observation retention and source representation. An ordinary structured-state baseline receiving the same facts and deterministic result would test whether EAL/2 contributes beyond information delivery and computation. A baseline receiving matched facts without the computed decision would test the added value of deterministic evaluation. Such comparisons require explicit, prospectively fixed inputs and endpoints.

The current synthetic cases should also be separated from a broader target population. Replication on new independently authored tasks, with external review of task realism and the reference decisions, would address common specification error. Independent coding should first validate a prospectively selected sample of the retained prose, with oversampling or complete review of ambiguous and disagreement-prone answers accounted for in the validation design. Only after the measurement procedure and nuisance estimates are adequate should a confirmatory allocation be frozen. The blocked planner should not be bypassed by filling ambiguous labels from the desired reference or recycling the pilot as a fresh evaluation.

6.3. Threats to validity

Construct validity. A three-way readiness category compresses the quality of an engineering assessment. The current outcome does not validate an explanation, action recommendation, source interpretation or assurance argument. Synthetic eligibility rules and one-minute snapshot validity are imposed contracts, and correct execution may preserve a poorly chosen rule. Token counts are a reproducible model resource measure, but omit development effort and much host cost.

Internal validity. Randomised paired ordering controls some order effects, but not every provider or concurrency effect. The EAL intervention differs in automatic evidence delivery as well as representation. The two model snapshots differ in more than reasoning effort. Tool use and answer length are outcomes of the assigned workflow and should not be adjusted for as if they were pretreatment covariates. AI coding was introduced after collection, can infer arm identity from answer wording and has no independent measured error rate.

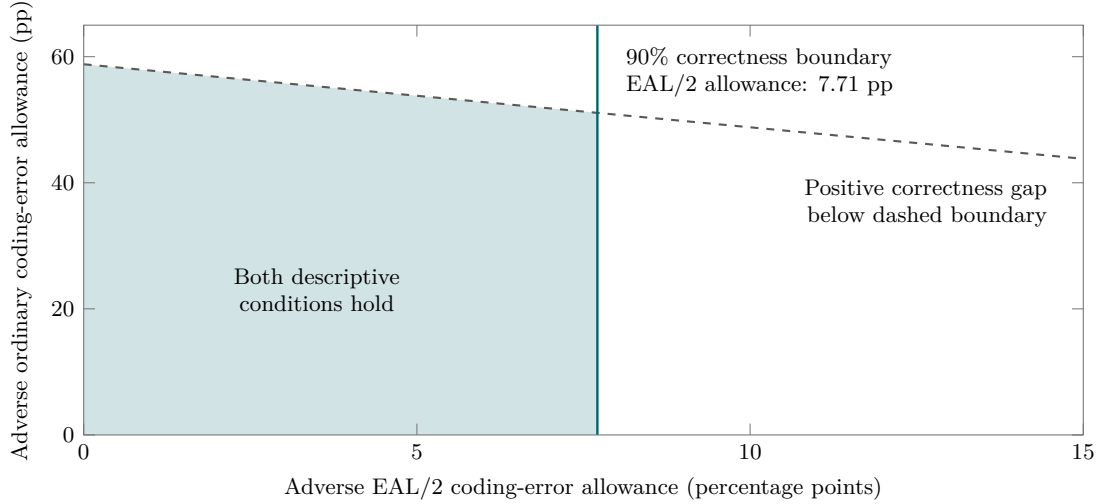


Figure 6: Assumption-based sensitivity of the descriptive correctness envelopes. The shaded region satisfies both a positive EAL/2–ordinary gap and an EAL/2 lower correctness bound above 90%, after adverse changes to resolved codes. Boundary points do not satisfy the strict inequalities. No assessor error rates have been measured. This is an exploratory algebraic calculation, not additional experimental data or a confidence region.

Statistical conclusion validity. The primary comparison preserves whole trajectories and all planned pairs. Sampling intervals nevertheless require independent repetitions and stable execution conditions; a common service batch can violate these assumptions. Four repeats per stratum are thin support for variance estimation. The resource interval is approximate. The post-collection diagnostic plots and coding sensitivity are exploratory, and there is no multiplicity-adjusted confirmatory claim over task or configuration subgroups.

External validity. Six authored cases, two snapshots from one provider and one collected batch do not establish performance over engineering projects, organisations or future models. No human participant study was conducted and no developer productivity outcome was measured. EAL source and host integrations were already supplied. A new adopter may incur substantial modelling, validation and maintenance costs before achieving comparable operation.

7. Replication and research transparency

The replacement `paper/` directory contains the article source, compiled PDF, include-ready TikZ and vector PDF figures, the syntax-highlighting definition, full retained EAL example, generated tables and an offline evidence snapshot [9]. The snapshot preserves complete annotated session rows, including initial prompts, final answers and evidence events. API calls are projected to usage, timing, identifiers and configuration fields; duplicated request text and provider response payloads are omitted from this compact package. The original archive and every copied source file are identified by SHA-256 hashes in its manifest.

Running `make reproduce` regenerates results and plot sources without network access or API credentials. Running `make check` verifies checksums, all reference decisions, all saved decision scores, pairing, allocation, attempt coverage and generated-output consistency, and parses the EAL example with the repository implementation. `make pdf` builds the figures and article; `make submission` produces a flat source bundle. The README specifies dependencies, command order, data provenance and what is and is not reproduced.

The source collection is revision-bound, and the compact snapshot prevents numerical reproduction from depending solely on a temporary workflow artefact. The original finished GitHub artefact is scheduled to expire on 27 December 2026. Repository access and a durable public archive must be arranged before a submission claims unrestricted public availability. The

package does not state that a journal has reviewed the work or that the current manuscript is accepted for publication.

8. Conclusion

The completed pilot supports a bounded descriptive finding: a host-assessed EAL/2 workflow preserves reference-matching decisions across fresh sessions more often than the implemented ordinary-notes workflow, while using 35.05% fewer cumulative model tokens. It does so with more recorded evidence acquisitions and with automatic delivery of checked context. The comparison therefore supports further evaluation of the workflow package, rather than an isolated effect of notation or a general model-capability claim.

The collection and offline processing are complete. Independent validation of the AI coding, resolution of the remaining ambiguous answers and a supported prospective allocation are not complete. Those limitations are retained in the data, plots and conclusion. The next scientific step is to validate the measurement and separate information delivery from representation and deterministic assessment in a prospectively specified comparison.

Declaration of generative AI use

OpenAI ChatGPT/Codex was used to assist task and software development, code the retained prose under the disclosed procedure, prepare this manuscript and write the plotting and reproduction code. The exact serving model used for coding was not exposed. Figures are programmatic renderings of retained observations or explicitly stated algebraic assumptions; they are not generated experimental observations. AI coding provenance and its absence of independent human validation are part of the method and limitations. This declaration does not assert that the present draft has completed human author review.

Appendix A. Uncertainty calculation and audit boundaries

For each paired block, the ten recipient outcomes give lower and upper sequence means, and their adverse cross-arm differences give $[l_b, u_b] \subseteq [-1, 1]$. The retained implementation bounds the sample variance over every completion of these intervals. With $m_b = (l_b + u_b)/2$, half-width $d_b = (u_b - l_b)/2$, sample variance s_m^2 and support width W , it uses

$$v_+ = \min \left\{ \left(\sqrt{s_m^2} + \sqrt{\frac{\sum_b d_b^2}{B-1}} \right)^2, \frac{BW^2}{4(B-1)} \right\}. \quad (\text{A.1})$$

The first term follows by applying the triangle inequality to centred observation vectors; the second bounds variance on the declared support. An empirical Bernstein radius is then

$$r = \sqrt{\frac{2v_+ \log(2/\alpha_t)}{B}} + \frac{7W \log(2/\alpha_t)}{3(B-1)}. \quad (\text{A.2})$$

The interval expands the mean lower and upper endpoints by r and clips to the support. Theorem 11 of Maurer and Pontil supplies the independent bounded-observation result [7]; the interval-completion envelope is an implementation choice. The two-sided three-endpoint display uses $\alpha_t = 0.05/6$ per tail. The one-sided joint decision uses 0.05 for each component. For absolute EAL/2 correctness the support is $[0, 1]$ and $W = 1$, whereas paired differences have $W = 2$.

The resource calculation works with paired cumulative input-plus-output counts and estimates the ratio of stratum-weighted means. It retains covariance within pairs and estimates within-stratum variance from the four repetitions, using a Satterthwaite-style effective degrees of freedom of 44.73 for the reported t correction. The article independently reconstructs its resource totals

and ratio; the approximate interval is retained from the revision-bound analysis implementation. The package labels this distinction instead of claiming that two implementations independently establish the interval.

The collection revision, offline-processing revision and paper analysis have different purposes. Collection determines which outputs were produced. Offline processing adds explicit AI annotation provenance and computes reports. Article processing verifies the compact evidence package and renders the manuscript’s tables and plots. The paper does not rewrite the frozen collection, its budget, run identity or original answer text.

References

- [1] L. Gao, A. Madaan, S. Zhou, U. Alon, P. Liu, Y. Yang, J. Callan, G. Neubig, PAL: Program-aided language models, in: Proceedings of the 40th International Conference on Machine Learning, Vol. 202 of Proceedings of Machine Learning Research, 2023, pp. 10764–10799. URL <https://proceedings.mlr.press/v202/gao23f.html>
- [2] L. Pan, A. Albalak, X. Wang, W. Wang, Logic-LM: Empowering large language models with symbolic solvers for faithful logical reasoning, in: Findings of the Association for Computational Linguistics: EMNLP 2023, 2023, pp. 3806–3824. doi:10.18653/v1/2023.findings-emnlp.248. URL <https://aclanthology.org/2023.findings-emnlp.248/>
- [3] S. Yao, J. Zhao, D. Yu, N. Du, I. Shafran, K. Narasimhan, Y. Cao, ReAct: Synergizing reasoning and acting in language models, ICLR 2023; arXiv:2210.03629 (2023). URL <https://arxiv.org/abs/2210.03629>
- [4] Object Management Group, Structured Assurance Case Metamodel (SACM), Version 2.3 (October 2023). URL <https://www.omg.org/spec/SACM/2.3>
- [5] N. F. Liu, K. Lin, J. Hewitt, A. Paranjape, M. Bevilacqua, F. Petroni, P. Liang, Lost in the middle: How language models use long contexts, Transactions of the Association for Computational Linguistics 12 (2024) 157–173. doi:10.1162/tacl_a_00638. URL <https://aclanthology.org/2024.tacl-1.9/>
- [6] L. Zheng, W.-L. Chiang, Y. Sheng, S. Zhuang, Z. Wu, Y. Zhuang, Z. Lin, Z. Li, D. Li, E. P. Xing, H. Zhang, J. E. Gonzalez, I. Stoica, Judging LLM-as-a-judge with MT-Bench and Chatbot Arena, NeurIPS 2023 Datasets and Benchmarks; arXiv:2306.05685 (2023). URL <https://arxiv.org/abs/2306.05685>
- [7] A. Maurer, M. Pontil, Empirical Bernstein bounds and sample variance penalization, COLT 2009; arXiv:0907.3740 (2009). URL <https://arxiv.org/abs/0907.3740>
- [8] A. Deng, U. Knoblich, J. Lu, Applying the delta method in metric analytics: A practical guide with novel ideas, KDD 2018; arXiv:1803.06336 (2018). doi:10.1145/3219819.3219919. URL <https://arxiv.org/abs/1803.06336>
- [9] EAL project, EAL/2 model-transfer pilot: retained collection, AI coding and offline analysis, collection 36477862213; processing 36492423435; fixed identities and checksums in the accompanying replication package (2026). URL <https://github.com/emmett08/earl>